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Exceptional degeneracies in traveling wave tubes with dispersive slow-wave structure including space-charge effect

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ABSTRACT

The interaction between a linear electron beam and a guided electromagnetic wave is studied in the context of exceptional points of degeneracy (EPD) supported by such an interactive system, focusing on a linear beam traveling wave tube (TWT) with a realistic helix waveguide slow-wave structure (SWS). The interaction is formulated by an analytical model that is a generalization of the Pierce model, assuming a one-dimensional electron flow along a dispersive single-mode guiding SWS and taking into account space-charge effects. The augmented model using phase velocity and characteristic impedance obtained via full-wave simulations is validated by calculating gain vs frequency and comparing it with that from more complex electron beam simulators. This comparison also shows the accuracy of our new model compared with respect to the nondispersive Pierce model. EPDs are then investigated using the augmented model, observing the coalescence of complex-valued wavenumbers and the system's eigenvectors. The point in the complex dispersion diagram at which the TWT-system starts/ceases to exhibit a convection instability, i.e., a mode starts/ceases to grow exponentially along the TWT, is the EPD. We also demonstrate the EPD existence by showing that the Puiseux fractional power series expansion well approximates the bifurcation of the dispersion diagram at the EPD. This latter concept also explains the "exceptional" sensitivity of the TWT-system to changes in the beam's electron velocity when operating near an EPD.

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High power traveling wave tube (TWT) amplifiers are of high importance for telecommunications, radar applications, including atmospheric studies, precision tracking, and high-resolution imaging.[1–3] In some occasion, metamaterials have been proposed to enhance the properties of electron beam devices.[4–7] A TWT uses a slow-wave structure (SWS) as a key component to harvest energy from an electron beam (e-beam) into radio frequency waves efficiently over broad bandwidths.[8,9] In this paper, we focus on a realistic helix TWT with dispersive SWS's characteristic parameters. The interaction with the e-beam affects the way electromagnetic (EM) waves propagate in the so-called "hot" circuit, i.e., accounting for the beam-electromagnetic mode interaction. We refer to the eigenmodes of the system where the e-beam interacts with the EM wave of the SWS as hot modes, which are the modes that retain properties of both EM and space-charge waves.

An exceptional point of degeneracy (EPD) in a system refers to the property of the system matrix that contains at least one nontrivial Jordan block structure, i.e., when two or more eigenvectors coalesce into a single degenerate one.[10–16] The concept of EPD has been studied in lossless, spatially,[17,18] or temporally[19,20] periodic structures, and in systems with loss and/or gain under parity-time symmetry.[13,21,22] We employ the Puiseux fractional power expansion series to illustrate the bifurcation of the system's dispersion diagram at the EPD.[23] The EPD has been studied for its applications in sensing devices[24,25] and oscillators.[26–29]

In Ref. 30, the EPD in a system of an e-beam interacting with an EM mode guided in a nondispersive SWS with distributed power extraction and without accounting for electrons' debunching is used to conceive an effective oscillator. The mathematical formulation in Ref. 31 does not include waveguide dispersion in the model, and above

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